

Lab1

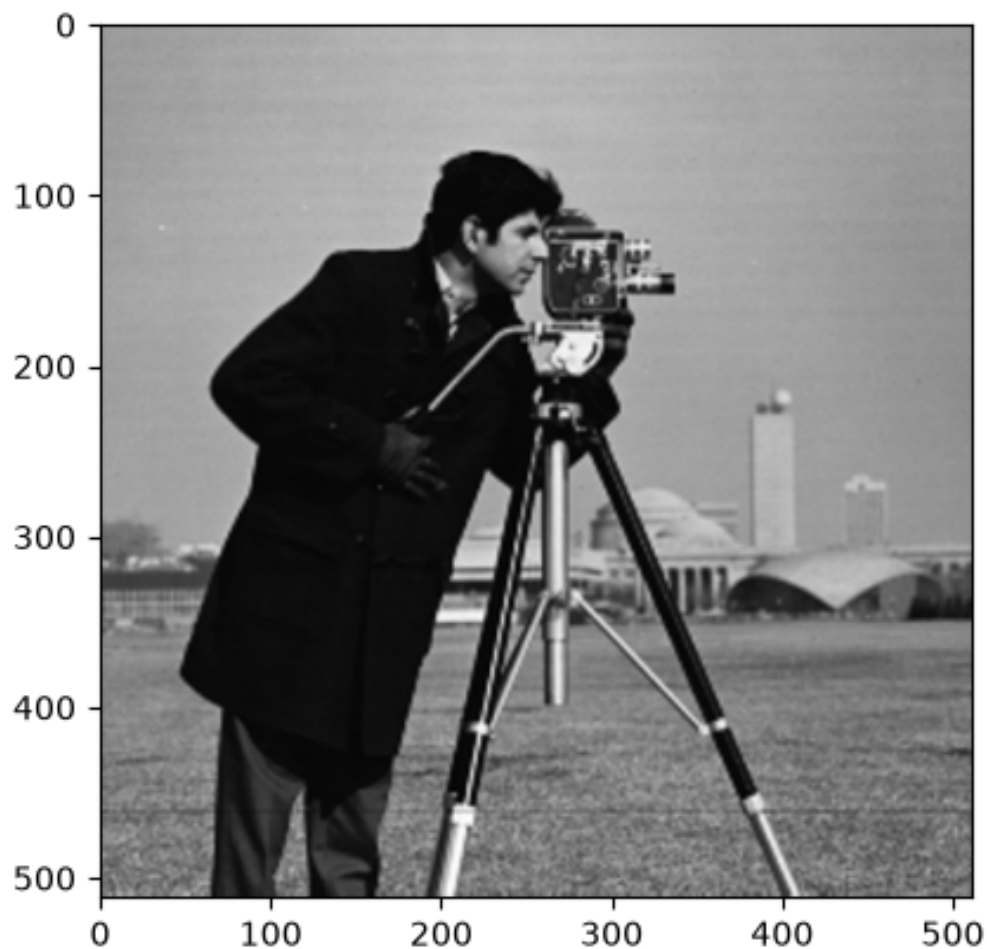
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Task 1 & 2:

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread('../images/cameraman.tif') #Load the image data into a NumPy array

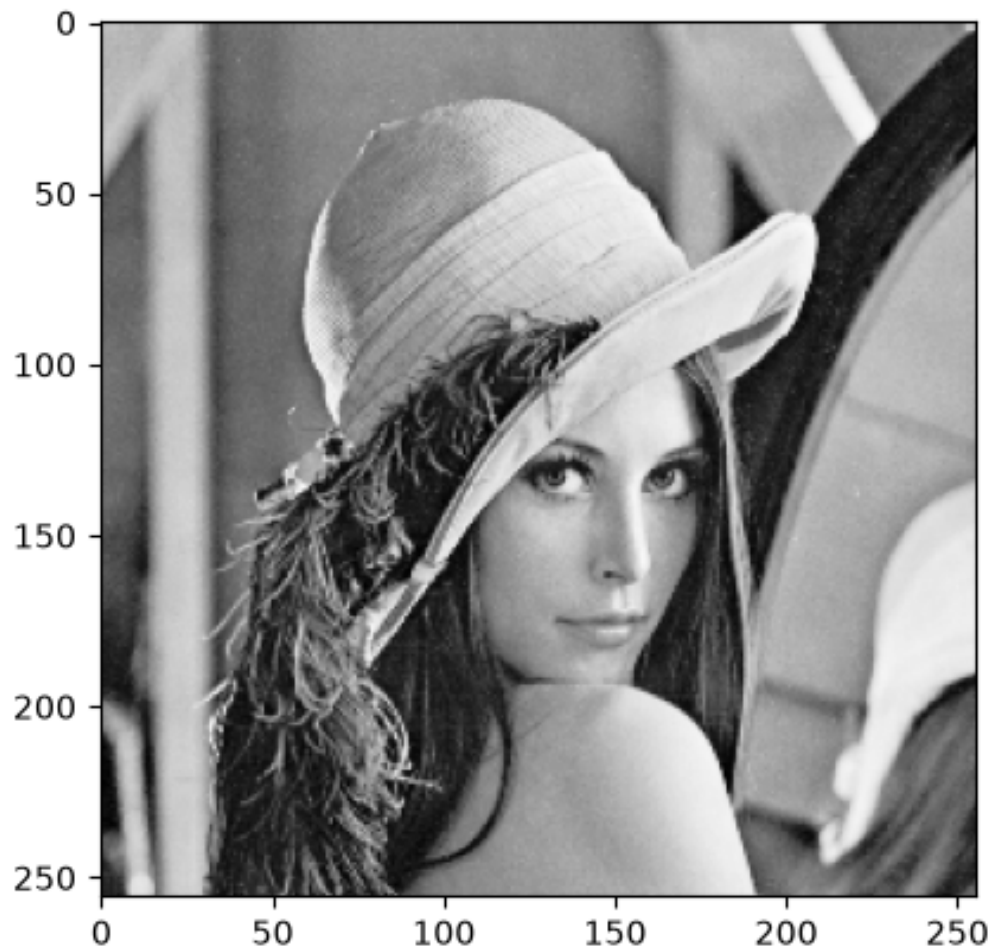
plt.imshow(img)
plt.show()
```



```
from PIL import Image
import matplotlib.cm as cm

img2 = Image.open('../images/lena_gray_256.tif') #load image into a PIL Image object

plt.imshow(img2, cmap=cm.Greys_r)
plt.show()
```



Task 3:

```
cv2.imwrite('new_image.jpg', img)
```

```
True
```

```
img2.save('new_image2.jpg')
```

```
print(img.shape) #Note: img1 loaded using open cv is a numpy array  
print(img)
```

```
(512, 512, 3)  
[[[156 156 156]  
  [157 157 157]  
  [160 160 160]  
  ...  
  [152 152 152]  
  [152 152 152]  
  [152 152 152]]  
  
 [[156 156 156]  
  [157 157 157]  
  [159 159 159]  
  ...  
  [152 152 152]  
  [152 152 152]  
  [152 152 152]]  
  
 [[158 158 158]  
  [157 157 157]  
  [156 156 156]]
```

```

...
[152 152 152]
[152 152 152]
[152 152 152]]

...

[[121 121 121]
 [123 123 123]
 [126 126 126]
 ...
 [121 121 121]
 [113 113 113]
 [111 111 111]]

[[121 121 121]
 [123 123 123]
 [126 126 126]
 ...
 [121 121 121]
 [113 113 113]
 [111 111 111]]

[[121 121 121]
 [123 123 123]
 [126 126 126]
 ...
 [121 121 121]
 [113 113 113]
 [111 111 111]]]

```

Task 4:

```

import numpy as np

# Convert the image to a NumPy array
img_array = np.array(img2)

# Print the array
print(img_array.shape)
print(img_array)

(256, 256)
[[162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 ...
 [ 48  57  54 ...  75  97  88]
 [ 46  50  52 ...  89 100  98]
 [ 44  55  54 ...  97 100 105]]

```